

Weekly Report

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Project: UAV-Based Traffic Safety & Video Analytics

Week: 28 Feb 2026 – 7 Mar 2026

Submission Date: 7 Mar 2026

Detailed Summary of Work Done During Current Week

A. Image Registration Validation

- Continued evaluation of the image registration pipeline developed in previous weeks.
- Tested registration stability across longer video sequences to analyze drift behavior.
- Verified frame alignment accuracy by visually comparing registered frames with the reference frame.
- Confirmed that the registration process maintains spatial consistency required for downstream trajectory analysis.

B. Zone Overlay Verification

- Verified that predefined polygon zones remain spatially consistent after frame registration.
- Conducted multiple tests across different video segments to ensure zones align correctly with road geometry.
- Confirmed that zone boundaries remain stable even in segments with noticeable UAV camera motion.

C. Semi-Automated Annotation Pipeline Development

Work this week continued on designing a scalable annotation workflow to manage large UAV datasets (~9000 frames).

1. Inference-Based Pre-Annotation

- Ran object detection inference on video frames to automatically generate bounding boxes.
- Extracted detection outputs including class labels, confidence scores, and bounding box coordinates.
- Organized detection outputs in a structured format for further processing.

2. CSV Annotation Data Structure

- Implemented CSV-based storage format for detections including:
 - Frame number
 - Bounding box coordinates
 - Class label
 - Confidence score
- This format provides an intermediate representation that can be converted into annotation files.

3. CSV to XML Conversion

- Continued development of a script to convert CSV detections into CVAT-compatible XML format.
- Ensured that the XML structure aligns with CVAT's video annotation schema.
- Performed initial tests importing generated XML annotations into CVAT.

4. CVAT-Based Annotation Refinement

- Tested loading of generated annotations in CVAT to evaluate usability for manual correction.
- Verified that automatically generated bounding boxes can be edited and adjusted efficiently.
- This workflow allows human annotators to refine predictions rather than manually labeling every frame.

D. Preliminary Trajectory Extraction

- Implemented centroid computation from bounding box coordinates.
- Generated initial trajectory points for selected vehicles in short video segments.
- Visualized vehicle paths across frames to verify trajectory extraction logic.
- Identified challenges related to vehicle identity consistency across frames.

Tasks Planned for Coming Week

A. Annotation Workflow Improvement

- Complete testing of CSV → XML conversion pipeline.
- Improve compatibility with CVAT video annotation format.
- Validate annotation workflow on larger frame segments.

B. Trajectory Development

- Extend trajectory extraction to longer video sequences.
- Improve consistency of vehicle identification across frames.
- Generate clearer trajectory visualizations for analysis.

C. Pipeline Integration

- Integrate detection, annotation, and trajectory extraction into a single processing workflow.
- Organize data outputs for easier analysis and documentation.